

# Bellman Optimality Equations, Value Iteration

Reinforcement Learning

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# 1 Understanding the Bellman Optimality Equations

We now explain carefully what the optimality equations mean and how the nonlinear Bellman operator should be interpreted.

## 1.1 The Optimality Equation

The Bellman optimality equation is

$$v(s) = \sup_{a \in A_s} \left\{ r(s, a) + \lambda \sum_{j \in S} P(j \mid s, a) v(j) \right\}, \quad s \in S. \quad (6.2.2)$$

This equation states:

The value of state  $s$  equals the best immediate reward plus the discounted expected future value, where the maximization is taken over all feasible actions at  $s$ .

Unlike policy evaluation, optimization is now embedded inside the recursion.

## 1.2 The Bellman Optimality Operator

Let  $V$  be the Banach space of bounded functions on  $S$  equipped with the sup norm.

Define the nonlinear operator  $\mathcal{L} : V \rightarrow V$  by

$$\mathcal{L}v = \sup_{d \in D^{MD}} \{r_d + \lambda P_d v\}. \quad (6.2.3)$$

Here:

- $D^{MD}$  is the set of deterministic Markov decision rules,
- $r_d(s) = r(s, d(s))$ ,
- $(P_d v)(s) = \sum_j P(j \mid s, d(s)) v(j)$ .

## 1.3 Componentwise Interpretation

The supremum in (6.2.3) is taken with respect to the *componentwise partial ordering* on  $V$ :

$$v \leq w \iff v(s) \leq w(s) \quad \forall s \in S.$$

Therefore,

$$(\mathcal{L}v)(s) = \sup_{d \in D^{MD}} [r_d(s) + \lambda (P_d v)(s)].$$

But since

$$r_d(s) + \lambda (P_d v)(s) = r(s, d(s)) + \lambda \sum_j P(j \mid s, d(s)) v(j),$$

the only part of  $d$  that affects the  $s$ -component is the action selected at state  $s$ .

Hence,

$$\sup_{d \in D^{MD}} [r_d(s) + \lambda(P_d v)(s)] = \sup_{a \in A_s} \left\{ r(s, a) + \lambda \sum_j P(j | s, a) v(j) \right\}.$$

Thus the Bellman optimality operator is defined globally, but its evaluation is performed state-wise.

#### 1.4 When the Supremum is Attained

If  $A_s$  is finite for every state, the supremum is attained and we may write

$$Lv = \max_{d \in D^{MD}} \{r_d + \lambda P_d v\}.$$

Equivalently,

$$(Lv)(s) = \max_{a \in A_s} \left\{ r(s, a) + \lambda \sum_j P(j | s, a) v(j) \right\}.$$

After computing the maximizing action for each state, we assemble these maximizing actions into a deterministic Markov policy.

#### 1.5 Conceptual Meaning

The Bellman optimality equation replaces the global problem

$$\sup_{\pi \in \Pi} \mathbb{E}^\pi \left[ \sum_{t=0}^{\infty} \lambda^t r(S_t, A_t) \right]$$

with a system of local maximization equations. This is the mathematical formulation of the *Principle of Optimality*:

An optimal strategy has the property that, whatever the initial state, the remaining decisions must constitute an optimal strategy with respect to the state resulting from the first decision.

## 2 Contraction of the Nonlinear Bellman Optimality Operator

We now prove that the nonlinear Bellman optimality operator is a contraction mapping on  $(V, \|\cdot\|_\infty)$  when  $0 \leq \lambda < 1$ .

Throughout, assume:

- $S$  is finite or countable,
- $A_s$  is finite for each  $s$  (so the maximum is attained),
- rewards are bounded,
- $0 \leq \lambda < 1$ .

## 2.1 Action-Value Backup

Given any  $v \in V$ , define the one-step lookahead **action-value function**

$$Q_v(s, a) := r(s, a) + \lambda \sum_{j \in S} P(j \mid s, a) v(j), \quad a \in A_s.$$

**Intuition.**  $Q_v(s, a)$  represents the total value obtained by taking action  $a$  at state  $s$  and then evaluating the future according to  $v$ .

## 2.2 Bellman Optimality Operator

Define the nonlinear Bellman optimality operator

$$(Lv)(s) := \max_{a \in A_s} Q_v(s, a) = \max_{a \in A_s} \left\{ r(s, a) + \lambda \sum_{j \in S} P(j \mid s, a) v(j) \right\}.$$

This operator performs one optimal Bellman backup.

## 2.3 Key Lemma: Max is 1-Lipschitz

**Lemma 2.1.** *Let  $\{x_a\}_{a \in \mathcal{A}}$  and  $\{y_a\}_{a \in \mathcal{A}}$  be real numbers indexed by the same finite set. Then*

$$\left| \max_{a \in \mathcal{A}} x_a - \max_{a \in \mathcal{A}} y_a \right| \leq \max_{a \in \mathcal{A}} |x_a - y_a|.$$

*Proof.* Let  $a_x \in \arg \max_a x_a$  and  $a_y \in \arg \max_a y_a$ . Then

$$\max_a x_a - \max_a y_a = x_{a_x} - y_{a_y} \leq x_{a_x} - y_{a_x} \leq \max_a |x_a - y_a|.$$

A symmetric argument bounds the reverse difference. □

**Intuition.** The max operator cannot create a difference larger than the largest elementwise difference already present.

## 2.4 Main Theorem: Contraction of $L$

**Theorem 2.2.** *For all  $v, w \in V$ ,*

$$\|Lv - Lw\|_\infty \leq \lambda \|v - w\|_\infty.$$

*Hence  $L$  is a contraction mapping on  $(V, \|\cdot\|_\infty)$  with modulus  $\lambda$ .*

*Proof.* Fix  $s \in S$ .

Define

$$Q_v(s, a) = r(s, a) + \lambda \sum_j P(j \mid s, a) v(j), \quad Q_w(s, a) = r(s, a) + \lambda \sum_j P(j \mid s, a) w(j).$$

Then

$$(Lv)(s) = \max_{a \in A_s} Q_v(s, a), \quad (Lw)(s) = \max_{a \in A_s} Q_w(s, a).$$

Using the previous lemma,

$$|(Lv)(s) - (Lw)(s)| \leq \max_{a \in A_s} |Q_v(s, a) - Q_w(s, a)|.$$

Now compute the difference:

$$Q_v(s, a) - Q_w(s, a) = \lambda \sum_j P(j | s, a) (v(j) - w(j)).$$

Taking absolute value,

$$|Q_v(s, a) - Q_w(s, a)| = \lambda \left| \sum_j P(j | s, a) (v(j) - w(j)) \right|.$$

Since  $P(\cdot | s, a)$  is a probability distribution,

$$\left| \sum_j P(j | s, a) (v(j) - w(j)) \right| \leq \sum_j P(j | s, a) |v(j) - w(j)| \leq \|v - w\|_\infty.$$

Therefore,

$$|Q_v(s, a) - Q_w(s, a)| \leq \lambda \|v - w\|_\infty.$$

Taking the maximum over  $a \in A_s$ ,

$$|(Lv)(s) - (Lw)(s)| \leq \lambda \|v - w\|_\infty.$$

Finally, take the supremum over all states:

$$\|Lv - Lw\|_\infty = \sup_{s \in S} |(Lv)(s) - (Lw)(s)| \leq \lambda \|v - w\|_\infty.$$

This completes the proof. □

### Conceptual Interpretation.

- The transition operator is an averaging operator and cannot increase the sup-norm error.
- The only scaling comes from the discount factor  $\lambda$ .
- The max operator is 1-Lipschitz in the sup norm and therefore preserves the contraction property.

### 3 Fixed Point Interpretation and Value Iteration

In Report 7 we proved the Banach fixed-point theorem: if  $(V, \|\cdot\|_\infty)$  is Banach and  $T : V \rightarrow V$  is a contraction, then  $T$  has a unique fixed point and repeated iteration converges to it.

In the previous section we proved that the Bellman optimality operator  $L : V \rightarrow V$  is a contraction with modulus  $\lambda \in [0, 1)$ . Therefore, all conclusions of the Banach fixed-point theorem apply directly to  $L$ .

#### 3.1 The Optimal Value Function as the Fixed Point of $L$

Recall the action-value backup for any  $v \in V$ :

$$Q_v(s, a) := r(s, a) + \lambda \sum_{j \in S} P(j \mid s, a) v(j), \quad a \in A_s.$$

The Bellman optimality operator is

$$(Lv)(s) := \max_{a \in A_s} Q_v(s, a).$$

**Theorem 3.1** (Unique fixed point of the Bellman optimality operator). *There exists a unique  $v^* \in V$  such that*

$$v^* = Lv^*.$$

*Moreover, for any initialization  $v^{(0)} \in V$ , the sequence defined by*

$$v^{(k+1)} = Lv^{(k)}$$

*converges in  $\|\cdot\|_\infty$  to  $v^*$ .*

*Proof.* We already proved  $L$  is a contraction on  $(V, \|\cdot\|_\infty)$  with modulus  $\lambda < 1$ . Since  $(V, \|\cdot\|_\infty)$  is Banach, the conclusion follows immediately from the Banach fixed-point theorem (Report 7).  $\square$

**Interpretation.** The optimality equations are exactly the fixed-point equation

$$v = Lv.$$

Thus, solving the discounted optimal control problem is equivalent to finding the unique fixed point of the nonlinear Bellman operator  $L$ .

#### 3.2 Value Iteration as Fixed-Point Iteration

Value Iteration is simply the fixed-point iteration

$$v^{(k+1)} = Lv^{(k)}.$$

Written componentwise, for each state  $s \in S$ ,

$$v^{(k+1)}(s) = \max_{a \in A_s} \left\{ r(s, a) + \lambda \sum_{j \in S} P(j \mid s, a) v^{(k)}(j) \right\}.$$

It is often convenient to compute the intermediate action-values

$$Q^{(k)}(s, a) := Q_{v^{(k)}}(s, a) = r(s, a) + \lambda \sum_{j \in S} P(j \mid s, a) v^{(k)}(j),$$

and then take

$$v^{(k+1)}(s) = \max_{a \in A_s} Q^{(k)}(s, a).$$

### 3.3 Value Iteration Algorithm (with explicit $Q$ -backup)

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**Algorithm 1:** Value Iteration via  $Q$ -Backups (Fixed-Point Iteration for  $v = Lv$ )

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**Input:** MDP model  $(S, \{A_s\}, P, r)$ , discount  $\lambda \in (0, 1)$ , tolerance  $\varepsilon > 0$ .

**Output:** Approximation  $v$  to the optimal value function  $v^*$ .

Initialize  $v^{(0)} \in V$  (e.g.  $v^{(0)}(s) = 0$  for all  $s$ );

```

for  $k = 0, 1, 2, \dots$  do
    foreach  $s \in S$  do
        foreach  $a \in A_s$  do
             $Q^{(k)}(s, a) \leftarrow r(s, a) + \lambda \sum_{j \in S} P(j \mid s, a) v^{(k)}(j);$ 
         $v^{(k+1)}(s) \leftarrow \max_{a \in A_s} Q^{(k)}(s, a);$ 
    if  $\|v^{(k+1)} - v^{(k)}\|_\infty < \varepsilon$  then
        return  $v^{(k+1)}$ ;

```

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**Optional: extracting a greedy policy.** Once  $v^{(k)}$  is close to  $v^*$ , we can define a greedy deterministic Markov rule

$$d^{(k)}(s) \in \arg \max_{a \in A_s} Q^{(k)}(s, a),$$

which approximates an optimal policy. When  $v^{(k)} \rightarrow v^*$ , any greedy selector  $d^{(k)}$  converges (up to ties) to an optimal stationary deterministic policy.